# Deep Reinforcement Learning for Anatomical Landmark Navigation in 3D Brain MRI

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## 1 Introduction

Localizing anatomical landmarks in brain MRI is a building block for atlas registration, surgical planning, and volumetric analysis. The standard approaches—template-based registration or supervised CNNs—both need a lot of manual annotation and can fail on atypical or pathological anatomy. This project takes a different angle: we frame landmark detection as a navigation problem where an RL agent starts at a random location in a 3D T1-weighted MNI152 volume and learns to navigate to a target subcortical structure. We're targeting 15 subcortical landmarks that cover a range of sizes, depths, and tissue contrasts (thalamus, hippocampus, lateral ventricles, brain-stem, putamen, etc.).

We built the system in-browser using TensorFlow.js for training and Niivue[1] for real-time neuroimaging visualization. The agent sees a $7 \times 7 \times 7$ voxel neighborhood around its current position plus a unit direction vector toward the target (346-dimensional state), and picks one of 6 actions (move $\pm 1$ voxel along $x$, $y$, or $z$). An episode ends when the agent gets within 3 voxels of the target (success) or runs out of 200 steps. So far we've implemented DQN and A2C and run initial experiments on 5 landmarks. DQN shows learning on most landmarks from unconstrained random starts, while A2C with a frozen MeshNet [6] convolutional backbone diverges even when starting within 50 voxels of the target.

## 2 Related Work

Ghesu et al. [1] were the first to frame anatomical landmark detection as RL navigation, using DQN in 3D CT with a fixed-scale patch observation and frame history buffer. Alansary et al. [2] extended this to MRI view planning with a multi-scale coarse-to-fine strategy. Both works only tested DQN variants (Double DQN, Dueling DQN) and found no single architecture wins across all landmarks. Our project differs in three ways: we compare across algorithm families (DQN vs. A2C/PPO), we use an explicit direction-to-target vector instead of frame history, and we plan systematic ablations of reward shaping and state representation. We also run entirely in-browser, using pretrained MeshNet [6] layers from Brainchop[2] as a frozen feature extractor for A2C.

[1] `https://github.com/niivue/niivue`
[2] `https://github.com/neuroneural/brainchop`

## 3 Approach

At each step the agent receives a $7 \times 7 \times 7$ neighborhood of min-max normalized voxel intensities (343 values) centered at its current position, concatenated with a 3D unit direction vector toward the target (346 dimensions total). It picks one of 6 actions: $\{+x, -x, +y, -y, +z, -z\}$, each moving one voxel. The reward has three components: a distance-based shaping signal $r_{\text{dist}} = -(d_{t+1} - d_t)$ (positive when moving closer), a step penalty of $-0.1$, and a $+10$ bonus for reaching within 3 voxels. Episodes cap at 200 steps.

**DQN** [4] takes the flat 346-dim state through a feedforward Q-network (256-128-64-6 dense layers, ReLU), with experience replay (10K buffer), epsilon-greedy exploration ($\epsilon$: $1.0 \to 0.05$, decay 0.995/episode), and a target network synced every 100 steps. **A2C** [5] was originally implemented with a frozen MeshNet backbone (two 3D conv layers, 30 filters, $3 \times 3 \times 3$ kernels, dilation 1 and 2) producing a 10,290-dim feature vector that was compressed through trainable $128 \to 64$ dense layers before splitting into policy and value heads. That variant diverged uniformly, so the results reported below use a flat A2C variant that takes the same 346-dim state as DQN through fully trainable dense layers ($128 \to 64$), with GAE ($\lambda = 0.95$), entropy regularization (0.01), and Adam (lr = 3e-4). **PPO** [3] uses the same flat 346-dim input and $128 \to 64$ shared trunk, a clipped surrogate objective (clip $\epsilon = 0.2$), GAE ($\lambda = 0.95$), 4 epochs per batch (batch size 64 trajectories), and Adam (lr = 3e-4). All three algorithms in the main comparison use a 50-voxel starting radius.

## 4 Experimental Results

We ran a full three-algorithm sweep across all 15 subcortical landmarks (DQN, A2C, PPO $\times$ 15 targets, 300 episodes each, 50-voxel starting radius, identical reward function, state representation, and 200-step cap). The A2C variant is the flat architecture (346-dim input, no frozen MeshNet backbone) described above. Performance is summarized using mean-over-first-50 (early) and mean-over-last-50 (late) episodes.

**Aggregate.** Averaged over 15 landmarks, PPO achieved the best late reward ($-18.7$), lowest late final distance (36.5 voxels), and highest late-window success rate (1.2%). Flat A2C improved slightly during training ($-43.2 \to -37.6$) but remained worse than PPO on nearly every landmark. DQN *regressed* on average ($-10.7 \to -32.6$, final distance 50.1): individual episodes early in training occasionally succeed under high-$\epsilon$ exploration, but the policy deteriorates as $\epsilon$ decays and the replay buffer is dominated by the agent's own drifting trajectories—a pattern consistent with Q-value overestimation/catastrophic forgetting.

**PPO** won 11/15 landmarks by late reward and 13/15 by final distance, and was the only algorithm with non-zero late-window success on a majority of landmarks (9/15). It did best on the cerebellar structures (Cerebellum-White-Matter: late $-7.1$; Cerebellum-Cortex: $-7.5$) and the 4th Ventricle ($-8.9$)—all spatially distinctive regions with clear tissue boundaries. Its hardest targets were the 3rd Ventricle ($-27.8$), Pallidum ($-23.3$), and Putamen ($-26.4$)—small, medial gray/CSF structures where the $7 \times 7 \times 7$ neighborhood captures less distinctive context.

**A2C** (flat) learned to navigate but under-performed PPO on every landmark except VentralDC (tie). It diverged on Hippocampus (late $-123$, dist 138) and Putamen ($-80$, dist 97)—small/curved structures where high-variance single-sample advantage updates appear fragile. The flat architecture clearly recovers the stability that the frozen MeshNet backbone destroyed in our earlier experiments, but pure A2C without PPO's trust-region clipping is still not robust.

**DQN** outperformed PPO only on Lateral-Ventricle (late $-15.6$ vs. $-18.5$), where it also hit the highest single-landmark success rate observed (4%). It was otherwise dominated by both on-policy

methods. Interestingly, DQN's early reward is the best of the three algorithms ($-10.7$)—random exploration under high $\epsilon$ plus dense distance shaping yields good initial trajectories—but the policy then regresses.

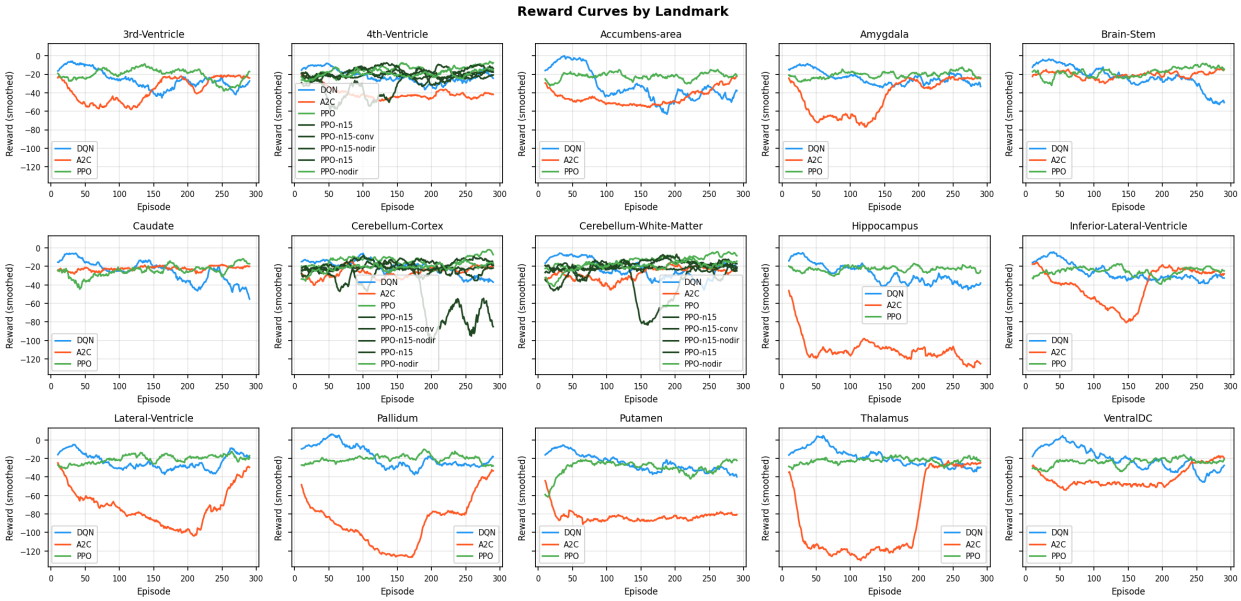


Figure 1: Smoothed mean episode reward over 300 training episodes for each algorithm on each of the 15 subcortical landmarks. PPO (green) improves or stays stable on all landmarks; A2C (orange) improves modestly but diverges on Hippocampus and Putamen; DQN (blue) regresses on most landmarks after initial high-$\epsilon$ exploration.

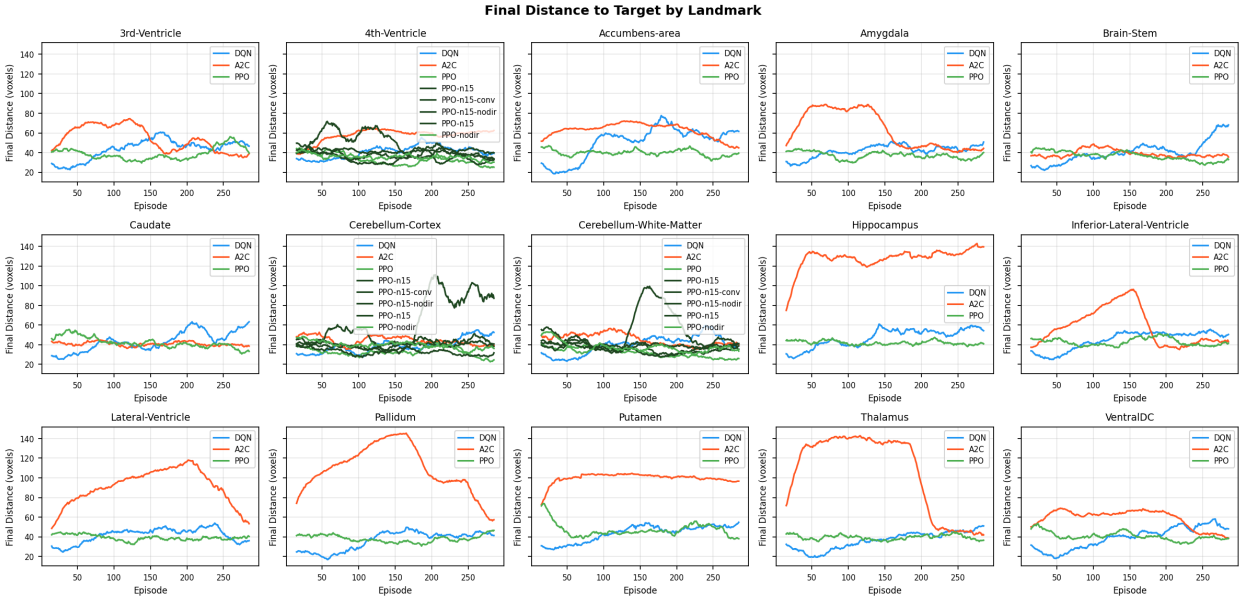


Figure 2: Smoothed mean final Euclidean distance to target (voxels), 30-episode rolling window. PPO achieves the lowest final distances on 13/15 landmarks.

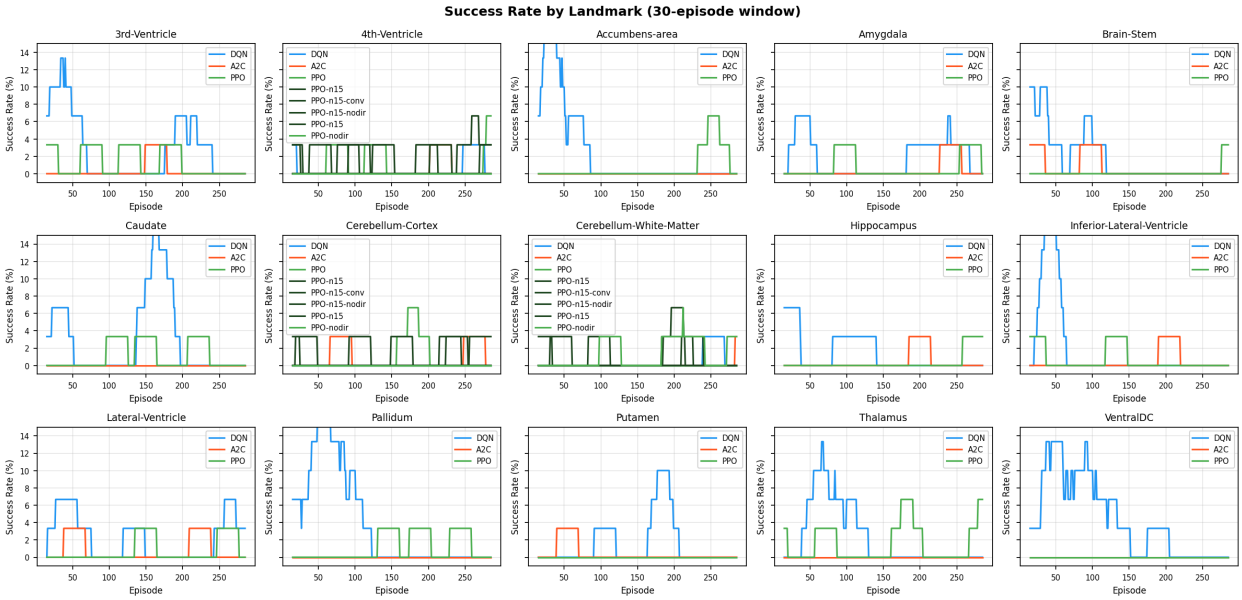


Figure 3: Success rate (within 3 voxels of target), 30-episode rolling window. Success rates remain in the low single digits across all algorithms at 300 episodes, but PPO is the only algorithm with non-zero late-window success on a majority of landmarks.

Table 1: Late-window (last 50 episodes) metrics per (algorithm, landmark). R = mean reward; d = mean final distance (voxels); % = success rate. All runs used 50-voxel starting radius and 200-step cap.

| Landmark | DQN R | A2C R | PPO R | DQN d | A2C d | PPO d | DQN % | A2C % | PPO % |
|---|---|---|---|---|---|---|---|---|---|
| 3rd-Ventricle | −32.0 | −23.3 | −27.8 | 47.8 | 37.8 | 45.9 | 0% | 0% | 0% |
| 4th-Ventricle | −22.7 | −43.3 | −8.9 | 41.6 | 62.4 | 25.8 | 2% | 0% | 4% |
| Accumbens-area | −40.6 | −27.9 | −20.0 | 59.1 | 46.7 | 37.4 | 0% | 0% | 2% |
| Amygdala | −27.4 | −24.8 | −19.2 | 46.8 | 43.1 | 36.8 | 2% | 0% | 2% |
| Brain-Stem | −46.2 | −16.2 | −12.1 | 63.9 | 35.8 | 31.1 | 0% | 0% | 2% |
| Caudate | −45.1 | −20.1 | −18.3 | 59.3 | 39.0 | 35.2 | 0% | 0% | 0% |
| Cerebellum-Cortex | −35.5 | −20.2 | −7.5 | 52.7 | 37.9 | 26.2 | 0% | 2% | 0% |
| Cerebellum-White-Matter | −21.3 | −23.5 | −7.1 | 41.5 | 40.5 | 25.7 | 2% | 2% | 0% |
| Hippocampus | −40.3 | −123.0 | −23.2 | 55.5 | 138.0 | 41.2 | 0% | 0% | 2% |
| Inferior-Lateral-Ventricle | −34.2 | −27.0 | −22.8 | 51.6 | 44.0 | 39.6 | 0% | 0% | 0% |
| Lateral-Ventricle | −15.6 | −40.7 | −18.5 | 35.3 | 60.7 | 38.7 | 4% | 0% | 2% |
| Pallidum | −24.7 | −46.4 | −23.3 | 41.6 | 65.0 | 43.0 | 0% | 0% | 0% |
| Putamen | −36.3 | −80.2 | −26.4 | 53.7 | 97.1 | 42.3 | 0% | 0% | 0% |
| Thalamus | −29.7 | −25.8 | −22.3 | 46.8 | 44.0 | 38.8 | 0% | 0% | 4% |
| VentralDC | −36.8 | −21.0 | −23.9 | 53.4 | 41.3 | 39.4 | 0% | 0% | 0% |
| **Mean** | **−32.6** | **−37.6** | **−18.7** | **50.1** | **55.5** | **36.5** | **0.7%** | **0.3%** | **1.2%** |

## Neighborhood-size ablation (PPO, $7^3$ vs $15^3$)

As a first ablation motivated by Ghesu et al.'s $25^3$ receptive field, we re-ran PPO on the three easiest landmarks (Cerebellum-White-Matter, Cerebellum-Cortex, 4th-Ventricle) with the observation cube enlarged from $7^3$ (346-dim state) to $15^3$ (3,378-dim state). Architecture, learning rate ($3 \times 10^{-4}$), entropy coefficient (0.01), and 300-episode budget held constant.

Table 2: PPO with $7^3$ vs $15^3$ observation window on the three easiest landmarks (last 50 episodes).

| Landmark | Late R ($7^3$) | Late R ($15^3$) | ΔR | Dist ($7^3$) | Dist ($15^3$) | Δdist |
|---|---|---|---|---|---|---|
| Cerebellum-White-Matter | −7.1 | −20.7 | −13.7 | 25.8 | 40.6 | +14.9 |
| Cerebellum-Cortex | −7.5 | −79.3 | −71.8 | 26.2 | 96.3 | +70.1 |
| 4th-Ventricle | −8.9 | −18.1 | −9.2 | 25.8 | 34.8 | +9.0 |

Across all three landmarks, enlarging the receptive field *hurt* performance under fixed hyperparameters. The first dense layer grew from ∼89K parameters (at $7^3$) to ∼865K (at $15^3$). Cerebellum-Cortex initially diverged to late reward −79 on one seed, but a second seed at the same configuration ran to −14.7, so the $15^3$-flat regression is noisy across seeds rather than systematically catastrophic. To separate receptive field from architecture-capacity, we then ran PPO at $15^3$ with a small trainable 3D conv trunk (two $3^3$ conv layers with 16 filters each, ReLU, max-pool 2, followed by the same 128/64 dense heads and direction concatenation).

Table 3: PPO at $15^3$ with flat MLP vs trainable 3D conv trunk, against the $7^3$ baseline (last 50 episodes, two independent seeds for the flat variant).

| Landmark | PPO $7^3$ flat | PPO $15^3$ flat (s1) | PPO $15^3$ flat (s2) | PPO $15^3$ conv |
|---|---|---|---|---|
| Cerebellum-White-Matter | −7.1 | −20.7 | −23.9 | −22.0 |
| Cerebellum-Cortex | −7.5 | −79.3 | −14.7 | −22.0 |
| 4th-Ventricle | −8.9 | −18.1 | −16.9 | −22.4 |

The trainable conv trunk lands inside the flat-seed noise band on all three landmarks and does not close the gap to the $7^3$ baseline. Giving the larger receptive field an appropriate inductive bias does not rescue it. A plausible explanation is that the explicit unit-direction-to-target vector already encodes the coarse navigation signal the agent needs; enlarging the voxel window then contributes little task-relevant information while requiring more training to fit.

### Direction-vector ablation

To test that hypothesis, we re-ran PPO with the 3-component direction input replaced by zeros (everything else held constant) on the three easiest landmarks, at both $7^3$ (flat) and $15^3$ (flat).

Table 4: Direction-vector ablation: PPO late-window reward (last 50 episodes) with direction input kept vs zeroed.

| Landmark | PPO $7^3$ kept | PPO $7^3$ zeroed | PPO $15^3$ kept (s1) | PPO $15^3$ zeroed |
|---|---|---|---|---|
| Cerebellum-White-Matter | −7.1 | −16.0 | −20.7 | −18.9 |
| Cerebellum-Cortex | −7.5 | −17.8 | −79.3 | −24.9 |
| 4th-Ventricle | −8.9 | −15.7 | −18.1 | −13.8 |

At $7^3$, zeroing the direction vector roughly doubles the negative late-window reward on all three landmarks, confirming that the unit-direction cue is doing most of the navigation work at the small receptive field. At $15^3$, the ablation is *not* worse than the direction-kept baseline on any landmark—and on Cerebellum-Cortex and 4th-Ventricle the zeroed variant is in fact closer to the $7^3$-kept reward than the $15^3$-kept variant was. That pattern is consistent with the direction vector acting as a shortcut that the flat $15^3$ trunk partially memorizes instead of learning from voxel context; removing the shortcut forces the larger window to be used, and the reward actually recovers. None of the $15^3$ runs reach the $7^3$-kept baseline, however, so the larger receptive field does not yet unlock navigation that the small window cannot also perform. A direction-zeroed run with the trainable conv trunk at $15^3$ is the next cell needed to finish this ablation grid.

Two follow-ups motivate the remaining work. First, absolute success rates are still low (single digits) even for PPO, confirming 300 episodes is short; a 1000–3000-episode PPO run on the three easiest landmarks will establish whether PPO plateaus or continues climbing. Second, the DQN regression pattern suggests the target-network sync interval (100 steps) and replay buffer size (10K) are under-tuned for navigation-length episodes; we plan a short DQN hyperparameter sweep rather than discarding the algorithm.

## 5 Future Milestones

- **April 14 (in progress):** Configurable observation neighborhood size ($5^3$–$25^3$), serialized into every result record so runs at different scales accumulate in the same corpus. Motivated by Ghesu et al., who use a $25^3$ ROI rather than our $7^3$.
- **April 17:** Neighborhood-size ablation on the three easiest PPO landmarks (Cerebellum-WM, Cerebellum-Cortex, 4th-Ventricle): sizes $7^3, 11^3, 15^3, 19^3$. For $19^3$+ we swap the flat-dense trunk for a small task-trained 3D conv (two $3^3$ conv layers, 16 filters, no frozen weights).
- **April 19:** Remaining ablations—reward formulation (sparse vs. dense vs. hybrid), with/without direction vector, starting-distance curriculum—only after the receptive-field ablation identifies a configuration with non-trivial success rates.

- **April 21:** Longer-training PPO (1000–3000 ep) on best neighborhood size; DQN hyperparameter sweep (replay buffer, target-sync interval).
- **April 22:** Difficulty ranking vs. anatomical properties; final report and presentation.
- **April 22–27:** Project presentations.

# 6 Conclusion

We've built a browser-based platform for deep RL landmark navigation in 3D brain MRI using TensorFlow.js and Niivue. A full three-algorithm comparison across all 15 subcortical landmarks ranks PPO first on 11/15 landmarks by late reward and 13/15 by final distance, with flat A2C a clear second and DQN third. PPO is the only algorithm that reaches non-zero late-window success on a majority of landmarks; A2C remains fragile on small structures (Hippocampus, Putamen); and DQN exhibits a regression pattern suggestive of Q-value instability as $\epsilon$ decays. Remaining work: extending PPO training to 1000–3000 episodes to establish a usable success-rate ceiling, a short DQN hyperparameter sweep (replay size, target-sync interval), and ablations of reward shaping, observation size, and the direction-vector component of the state before project presentations on April 22–27.

# References


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